1.GUI

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/textView1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="MOBILE APPLICATION"
        android:textSize="24sp"
        android:textStyle="bold" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Font Size" />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Colour" />

</LinearLayout>
```

MainActivity.java

```
package com.example.gui;

import android.app.*;
```

```
import android.os.*;
import android.graphics.Color;
import android.widget.*;

public class MainActivity extends Activity {
    float f=24;
    int i=0;
    int[] c={Color.BLUE,Color.GREEN,Color.RED,Color.BLACK};

    public void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        TextView t=findViewById(R.id.textView1);

        findViewById(R.id.button1).setOnClickListener(v -> {
            f+=4;
            if(f>40) f=20;
            t.setTextSize(f);
        });

        findViewById(R.id.button2).setOnClickListener(v -> {
            t.setTextColor(c[i++]);
            if(i==4) i=0;
        });
    }
}
```

2. LAYOUT

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
android:orientation="vertical"
android:padding="20dp">

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Student Information"
    android:textSize="24sp" />

<EditText
    android:id="@+id/n"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Name" />

<EditText
    android:id="@+id/d"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="D.No" />

<EditText
    android:id="@+id/de"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Department" />

<EditText
    android:id="@+id/c"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Contact Number"
    android:inputType="phone" />

<Button
    android:id="@+id/b"
    android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="OK" />

</LinearLayout>
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/t"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20sp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.layout;

import android.app.*;
import android.os.*;
import android.content.*;
import android.widget.*;

public class MainActivity extends Activity {
    public void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        EditText n=findViewById(R.id.n), d=findViewById(R.id.d),
```

```
            de=findViewById(R.id.de), c=findViewById(R.id.c);

        findViewById(R.id.b).setOnClickListener(v -> {
            Intent i=new Intent(this, MainActivity2.class);
            i.putExtra("n", n.getText().toString());
            i.putExtra("d", d.getText().toString());
            i.putExtra("de", de.getText().toString());
            i.putExtra("c", c.getText().toString());
            startActivity(i);
        });
    }
}
```

MainActivity2.java

```
package com.example.layout;
import android.app.*;
import android.content.Intent;
import android.os.*;
import android.widget.*;

public class MainActivity2 extends Activity {
    public void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main2);

        TextView t=findViewById(R.id.t);
        Intent i=getIntent();

        t.setText("Name: "+i.getStringExtra("n")
                +"\nD.No: "+i.getStringExtra("d")
                +"\nDepartment: "+i.getStringExtra("de")
                +"\nContact: "+i.getStringExtra("c"));
    }
}
```

AndroidManifest.xml:

(add after </activity> in default AndroidManifest.xml code)

<activity android:name=".MainActivity2" />

3. CALCULATOR

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="15dp">

    <EditText
        android:id="@+id/firstNumber"
        android:hint="First Number"
        android:inputType="numberDecimal|numberSigned"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/secondNumber"
        android:hint="Second Number"
        android:inputType="numberDecimal|numberSigned"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/add"
        android:text="Addition"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/sub"
        android:text="Subtraction"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/result"
        android:textSize="24sp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.calculator;

import android.app.*;
import android.os.*;
import android.widget.*;

public class MainActivity extends Activity {
    public void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        EditText firstNumber = findViewById(R.id.firstNumber);
        EditText secondNumber = findViewById(R.id.secondNumber);
        TextView result = findViewById(R.id.result);

        findViewById(R.id.add).setOnClickListener(v -> {
            double a = Double.parseDouble(firstNumber.getText().toString());
            double b1 =
Double.parseDouble(secondNumber.getText().toString());
            result.setText("Result: " + (a + b1));
        });

        findViewById(R.id.sub).setOnClickListener(v -> {
            double a = Double.parseDouble(firstNumber.getText().toString());
            double b1 =
Double.parseDouble(secondNumber.getText().toString());
```

```
            result.setText("Result: " + (a - b1));
        });
    }
}
```

4. GRAPHICAL LINE

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:id="@+id/imageView1"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</RelativeLayout>
```

MainActivity.java

```
package com.example.graphical;

import android.app.*;
import android.os.*;
import android.graphics.*;
import android.view.*;
import android.widget.*;

public class MainActivity extends Activity {
    Bitmap b;
    Canvas c;
    Paint p = new Paint();
    ImageView img;
    float x, y;

    public void onCreate(Bundle s) {
```

```
        super.onCreate(s);
        setContentView(R.layout.activity_main);
        img = findViewById(R.id.imageView1);
        b = Bitmap.createBitmap(800, 1200, Bitmap.Config.ARGB_8888);
        c = new Canvas(b);
        p.setColor(Color.BLUE);
        img.setImageBitmap(b);

        img.setOnTouchListener((v, e) -> {
            if (e.getAction() == 0) {
                x = e.getX();
                y = e.getY();
            } else if (e.getAction() == 1) {
                c.drawLine(x, y, e.getX(), e.getY(), p);
                img.invalidate();
            }
            return true;
        });
    }
}
```

5. DATABASE

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">

    <EditText
        android:id="@+id/name"
        android:hint="Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
```

```
<EditText
    android:id="@+id/regno"
    android:hint="Register Number"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<EditText
    android:id="@+id/dept"
    android:hint="Department"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:id="@+id/add"
    android:text="Add"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:id="@+id/view"
    android:text="View"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:id="@+id/update"
    android:text="Update"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>

<Button
    android:id="@+id/delete"
    android:text="Delete"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"/>
```

```
    <TextView
        android:id="@+id/output"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.database;

import android.app.*;
import android.os.*;
import android.database.*;
import android.database.sqlite.*;
import android.widget.*;

public class MainActivity extends Activity {
    EditText name, regno, dept;
    TextView output;
    SQLiteDatabase db;

    public void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        name = findViewById(R.id.name);
        regno = findViewById(R.id.regno);
        dept = findViewById(R.id.dept);
        output = findViewById(R.id.output);

        db = openOrCreateDatabase("StudentDB", 0, null);
        db.execSQL("CREATE TABLE IF NOT EXISTS students(reg TEXT
PRIMARY KEY,name TEXT,dept TEXT)");

        findViewById(R.id.add).setOnClickListener(v -> {
            db.execSQL("INSERT OR REPLACE INTO students
VALUES(?,?,?)",
```

```
                new String[]{regno.getText().toString(),
                        name.getText().toString(), dept.getText().toString()});
            msg("Added");
        });

        findViewById(R.id.update).setOnClickListener(v -> {
            db.execSQL("UPDATE students SET name=?,dept=? WHERE
reg=?",
                new String[]{name.getText().toString(),
                        dept.getText().toString(), regno.getText().toString()});
            msg("Updated");
        });

        findViewById(R.id.delete).setOnClickListener(v -> {
            db.delete("students", "reg=?",
                    new String[]{regno.getText().toString()});
            msg("Deleted");
        });

        findViewById(R.id.view).setOnClickListener(v -> {
            Cursor c = db.rawQuery("SELECT * FROM students", null);
            String s = "";
            while (c.moveToNext())
                s += "Register Number: " + c.getString(0) + "\nName: " +
c.getString(1)
                        + "\nDepartment: " + c.getString(2) + "\n";
            c.close();
            output.setText(s.isEmpty() ? "No records" : s);
        });
    }

    void msg(String s) {
        Toast.makeText(this, s, Toast.LENGTH_SHORT).show();
    }
}
```